

Classes and Objects Questions

Q1. Create a class Student with data members: name and marks. Use a constructor to initialize the values and display the student details using a member function.

Q2. Design a class Car that has attributes: brand and price. Write a constructor to assign values and create two objects to display the details of both cars.

Q3. Write a C++ program to create a class Rectangle with length and breadth as data members. Use a constructor to initialize the dimensions and calculate the area of the rectangle.

Q4. Create a class Employee with data members: emp_id and salary. Initialize the values using a parameterized constructor and display the employee information.

Q5. Define a class BankAccount with account number and balance. Use a constructor to set the initial balance and write a function to display account details.

Q6. Create a class Book that stores book title and price. Initialize the values using a constructor and create multiple objects to show how each object stores different data. Also explain the role of the constructor in this program.

Q7. A furniture showroom wants to develop a simple C++ program to manage details of different types of tables. Each table can be purchased in different ways:

- Some customers buy a table by specifying only the price (default table)
- Some customers specify the material and price
- Some customers specify material, price, and number of legs (custom table)

Design a class Table that uses **constructor overloading** to handle all three cases:

1. A default constructor that sets material as "Wood", price as 5000, and legs as 4
2. A constructor that accepts material and price
3. A constructor that accepts material, price, and number of legs

Create three objects using different constructors and display the details of each table.

Q.8 Write a C++ program to demonstrate **function overloading** by creating multiple functions named volume:

- One function to calculate the volume of a cube (side)
- One function to calculate the volume of a cuboid (length, breadth, height)
- One function to calculate the volume of a cylinder (radius, height)

Call all the overloaded volume functions in the main() function and display the calculated volumes.

Q.9 Electricity Billing System (Function Overloading)

An electricity board wants to design a billing system that calculates electricity charges for different types of consumers. The billing rules are:

- If only **units consumed** are provided → calculate bill at ₹5 per unit.
- If **units + rate per unit** are provided → calculate accordingly.
- If **units + rate + fixed charge** are provided → add fixed charge to total bill.

Requirements:

1. Create a class ElectricityBill.
2. Implement **function overloading** for a function named calculateBill():
 - I. calculateBill(int units)
 - II. calculateBill(int units, float rate)
 - III. calculateBill(int units, float rate, float fixedCharge)
3. Display the calculated bill in each case.
4. In main(): Call all three overloaded functions using different arguments.

Q.10 Online Shopping Cart

An e-commerce website wants to represent items in a user's shopping cart.

Each item should contain:

- Product name
- Price
- Quantity

Questions

1. Create a class called **Product**.
2. Use a **constructor** to initialize product details.
3. Write a function to calculate **total cost = price × quantity**.
4. Create two objects for different products.
5. When the program ends, the **destructor should display a message** indicating that the product is removed from memory.

Q.11. An online learning platform wants to store course information.

Each course should contain:

- Course name
- Instructor name
- Course duration (weeks)

Questions

1. Create a class Course.
2. Initialize course details using a constructor.
3. Write a function to display course details.
4. Create two course objects.
5. Add a destructor that prints a message when a course object is deleted.

Q12. A mobile store wants to maintain information about different mobile phones. Sometimes the store knows **only the brand**, sometimes it knows **brand and price**, and sometimes it knows **brand, price, and storage capacity**.

Problem Statement

Design a **class Mobile** that stores the following information:

- Brand
- Price
- Storage (in GB)

Requirements

1. Implement **constructor overloading** with the following constructors:
 - A constructor that initializes **only the brand**.
 - A constructor that initializes **brand and price**.
 - A constructor that initializes **brand, price, and storage**.
2. Write a **display function** to print the mobile details.
3. Create **three objects** using different constructors.
4. Display the details of each mobile.

Q.13. A computer shop wants to store information about laptops in its inventory. Sometimes the shop knows only the brand, sometimes brand and RAM, and sometimes brand, RAM, and price.

Requirements

1. Create a class Laptop with the following data members:
 - Brand
 - RAM
 - Price
2. Implement function overloading using a function called setDetails():
 - setDetails(string brand): sets only the brand
 - setDetails(string brand, int ram): sets brand and RAM
 - setDetails(string brand, int ram, int price): sets brand, RAM, and price
3. Create another function display() to show the laptop details.
4. In the main() function:
 - Create three laptop objects.
 - Call different overloaded setDetails() functions for each object.
 - Display the laptop information.

Q.14 A transport department wants to register vehicles with different levels of information. Each vehicle stores: Vehicle Number, Owner Name, Vehicle Type and Registration Fee

Requirements:

1. Create a class Vehicle.
2. Implement **constructor overloading**:

Default Constructor

- Initializes vehicle number as "Not Assigned"
- Owner name as "Unknown"
- Type as "General"
- Fee as 0

Parameterized Constructor (2 parameters): Initializes vehicle number and owner name.

Parameterized Constructor (4 parameters): Initializes all details.

- Create a function displayVehicle() to show vehicle details.
- In main(): Create three objects using different constructors and display details of all vehicles.